



Lab Report

CDS3006 – Data Analysis and Visualization

Class Id: BL2025260500028

Slot: A14+E14

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Submitted by

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Submitted to

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List of Experiments

[illegible]

Experiment No: 1 Quality Analysis of Ramen Ratings Using Stacked Bar Chart

Dt: 14/03/2026

Aim: Conduct a quality analysis on sample data set of ramen ratings using tableau Stacked Bar Chart.

Requirements:

- Tableau Desktop
- Operating System: Windows /macOS
- Dataset file: ramen-ratings.csv

Step by Step Implementation:

Step 1: Launch Tableau

Open Tableau on the system to begin the visualization process.

Step 2: Load the Dataset

1. Click File then Open.
2. Select the dataset ramen-ratings.csv.

Step 3: Verify Data Types

Check the data types of the attributes:

- Brand, Country, Style, Variety → Dimension
- Stars → Measure (numeric rating)

Step 4: Create Quality Categories

To analyse ramen quality, create rating categories.

1. Right-click in the Data Pane.
2. Click Create → Calculated Field.
3. Name the field Quality Category.

Use the formula:

IF [Stars] >= 4 THEN "High Quality"

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ELSEIF [Stars] >= 2.5 THEN "Medium Quality"
ELSE "Low Quality"
END
```

Step 5: Create the Bar Chart

1. Open Sheet 1.
2. Drag Country to the Rows shelf.
3. Drag Brand to the Columns shelf.
4. Change Brand to Measure → Count to show the number of ramen products.

Step 6: Create the Stacked Bar Chart

1. Drag Quality Category to the Color option in the Marks Card.

The bars will now be divided into High, Medium, and Low quality segments, forming a stacked bar chart.

Step 7: Add Data Labels

1. Drag COUNT(Brand) to Label in the Marks Card.
2. Enable Show Mark Labels to display values on the chart.

Step 8: Show Percentage Distribution

1. Right-click COUNT(Brand).
2. Select Quick Table Calculation → Percent of Total.

This displays the percentage of each quality category within every country.

Step 9: Improve Visualization

1. Sort Country by number of ramen products.
2. Add the chart title: “Quality Analysis of Ramen Ratings using Stacked Bar Chart.”
- 3.

Step 10: Analyse the Results

The stacked bar chart helps in:

- Comparing ramen quality distribution across countries.

- Identifying countries with higher rated ramen products.
- Displays quality category, country and count of rating.

Output:

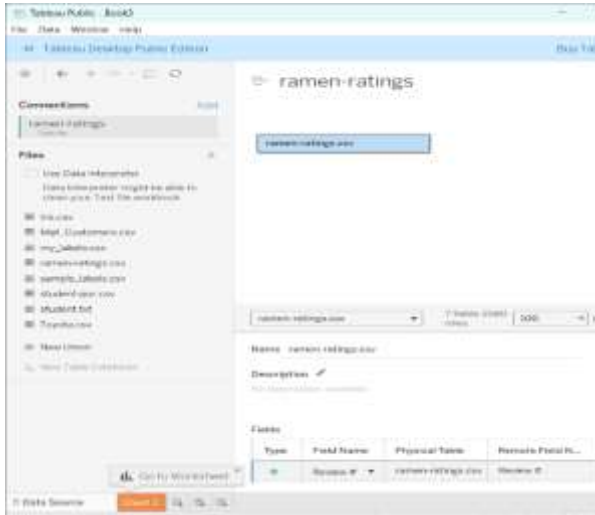


Fig1: Tableau is launched and dataset is loaded.

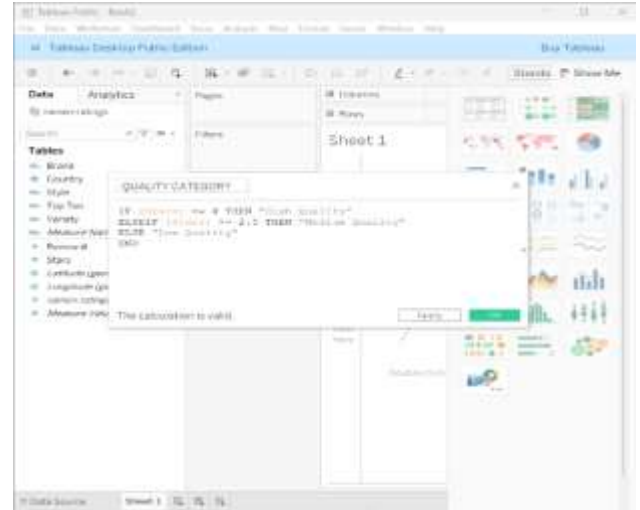


Fig2 : Created Quality Categories

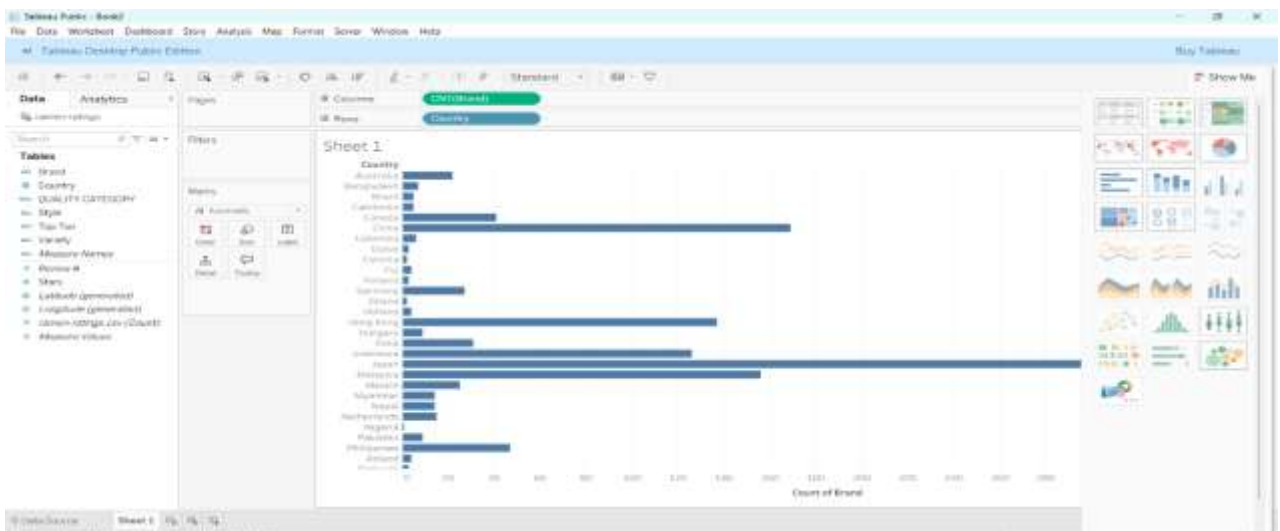


Fig 3: Created Bar Chart with Brand as Column and Country as Row.

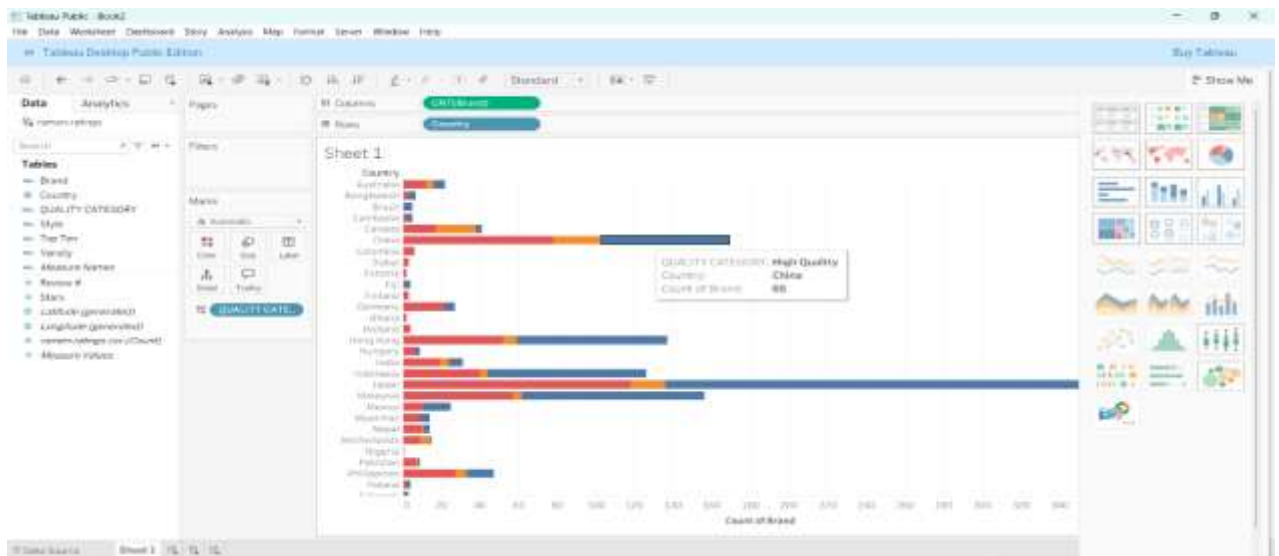


Fig 4: Created a Stacked Bar Chart and divided into High, Medium, and Low quality segments.

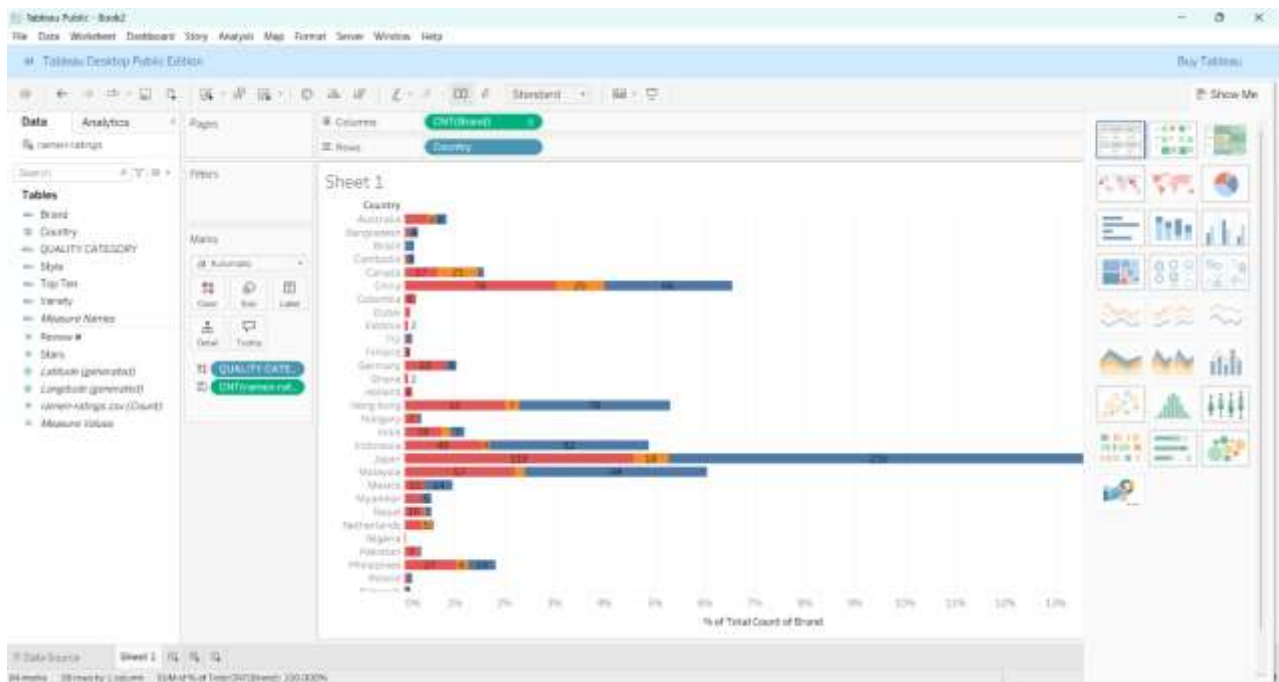


Fig 5: Displays the percentage of each quality category within every country.

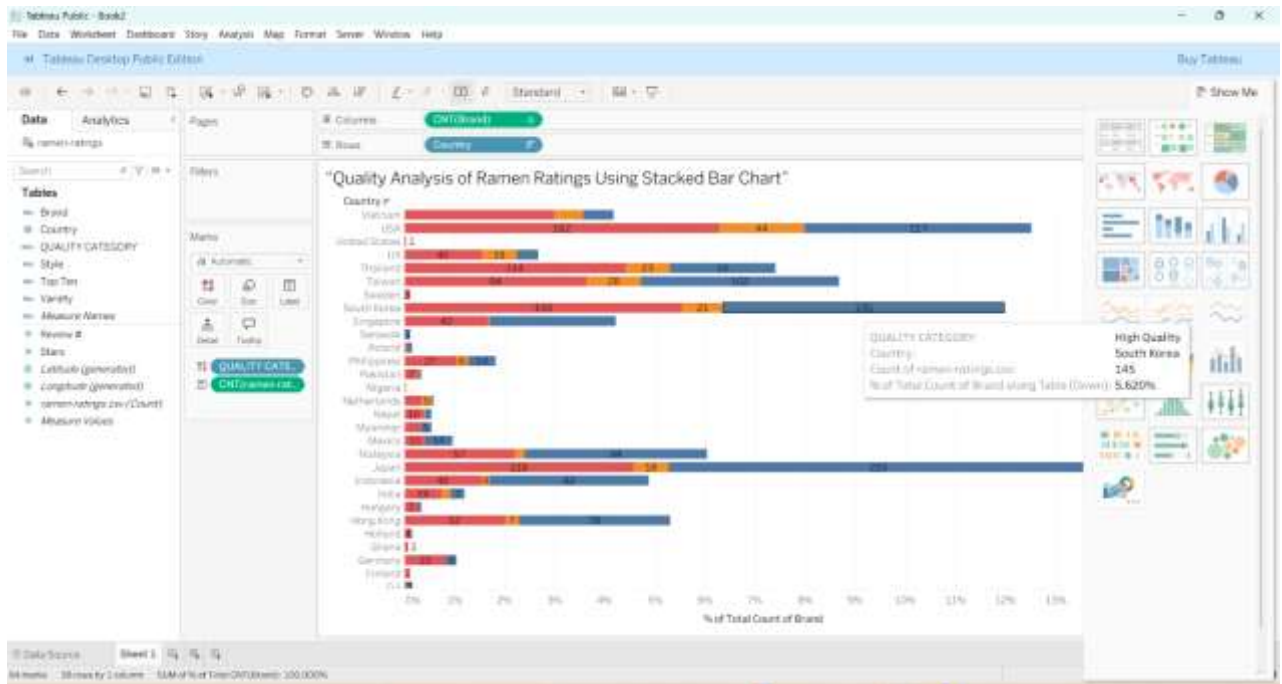


Fig 6: Sorted and added a title to get the final result.

Inference:

- The stacked bar chart successfully displayed the quality distribution of ramen products based on ratings.
- Countries with larger high-quality segments indicate better rated ramen products.
- Some countries show a higher proportion of medium or low-rated products, indicating variation in product quality.
- The visualization made it easier to compare the quality levels of ramen products across different regions.

Submitted by

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Experiment No: 2

Dt: 14/03/2026

Aim: Pattern analysis on sample data set using histogram on tableau

Requirements:

- Tableau Desktop / Tableau Public
- Operating System: Windows / macOS
- Dataset file: ramen-ratings.csv

Step by Step Implementation:

Step 1: Open Tableau and load the dataset ramen-ratings.csv using Connect → Text File.

Step 2: Check the Stars column and remove *Unrated* values if present so that it remains a numeric measure.

Step 3: Right-click Stars → Create → Bins, and set an appropriate bin size (for example 0.5 or 1).

Step 4: Drag Stars (bin) to Columns and drag COUNT(Brand) (or count of records) to Rows.

Step 5: Ensure the Marks type is Bar to display the histogram.

Output:

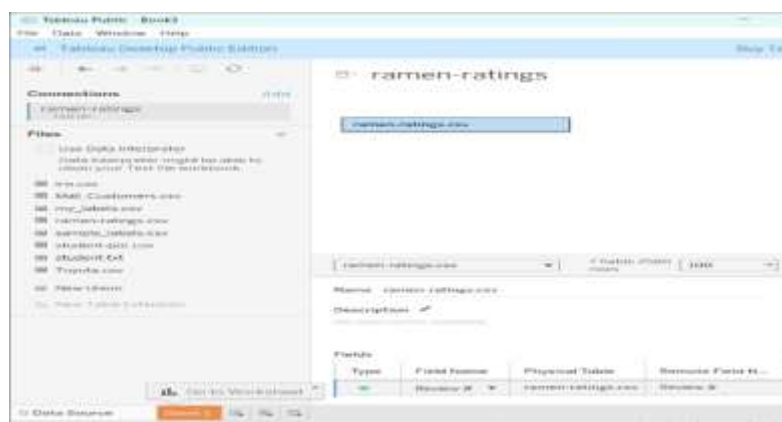


Fig1: Tableau is launched and dataset is loaded.



Fig 2: Size of bin for stars is assigned as 0.5

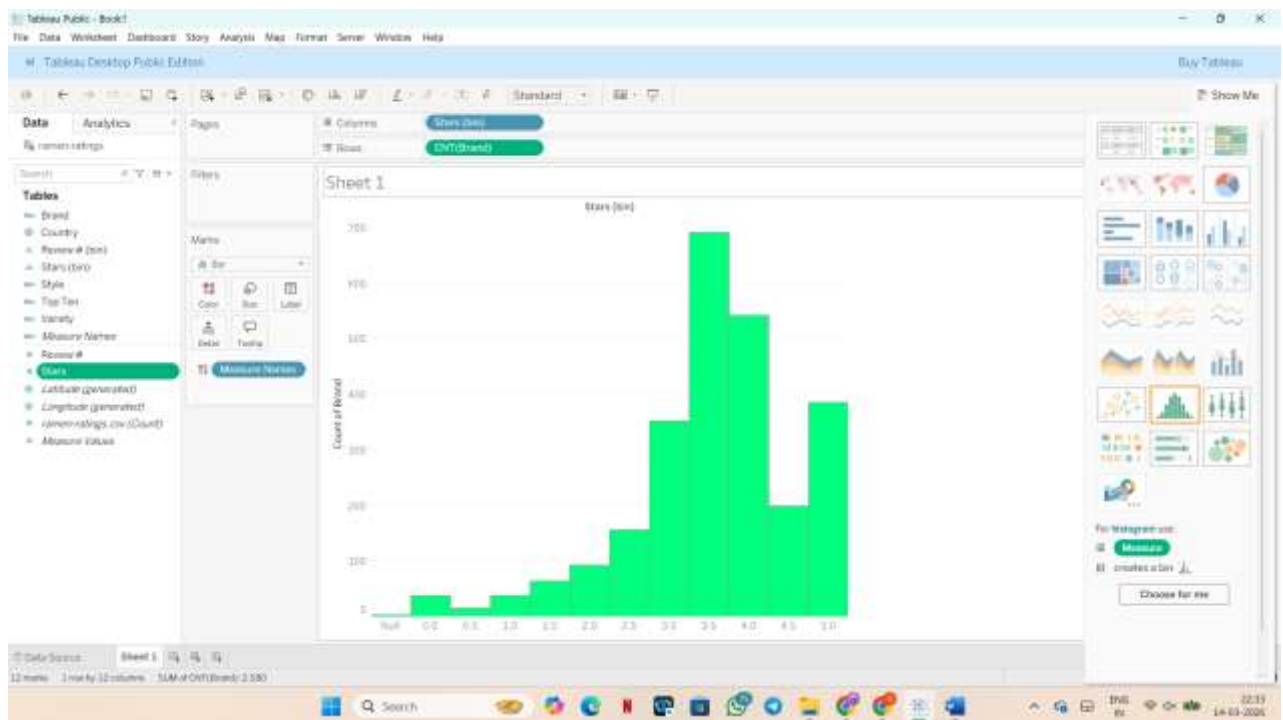


Fig 3: Histogram is created by assigning col and row as stars(bin) and Brand respectively.

Inference:

The histogram visualization created in Tableau shows the distribution pattern of ramen ratings in the dataset. The bars represent the frequency of ratings within different rating ranges (bins). From the histogram, it can be observed that most ramen products fall within certain rating ranges, indicating common quality levels. This visualization helps in identifying patterns, concentration of ratings, and the overall spread of the data.

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Experiment No: 3

Dt: 14/03/2026

Aim: Comparative analysis of two data sets using Butterfly chart on Tableau

Requirements:

- Tableau Desktop
- Operating System: Windows /macOS
- Dataset file: ramen-ratings.csv

Step by Step Implementation:

Step 1: Open Tableau and load the dataset ramen-ratings.csv using Connect → Text File.

Step 2: Drag Country → Rows to display categories for comparison.

Step 3: Drag AVG(Stars) → Columns to create the first bar chart representing average ramen ratings by country.

Step 4: Create a calculated field named Negative Count using the formula:

`-COUNT([Brand])`

Step 5: Drag Negative Count → Columns to create bars on the opposite side, forming a Butterfly Chart for comparison.

Step 6: Assign different colors to both measures and add labels to clearly visualize the comparison.

Output:

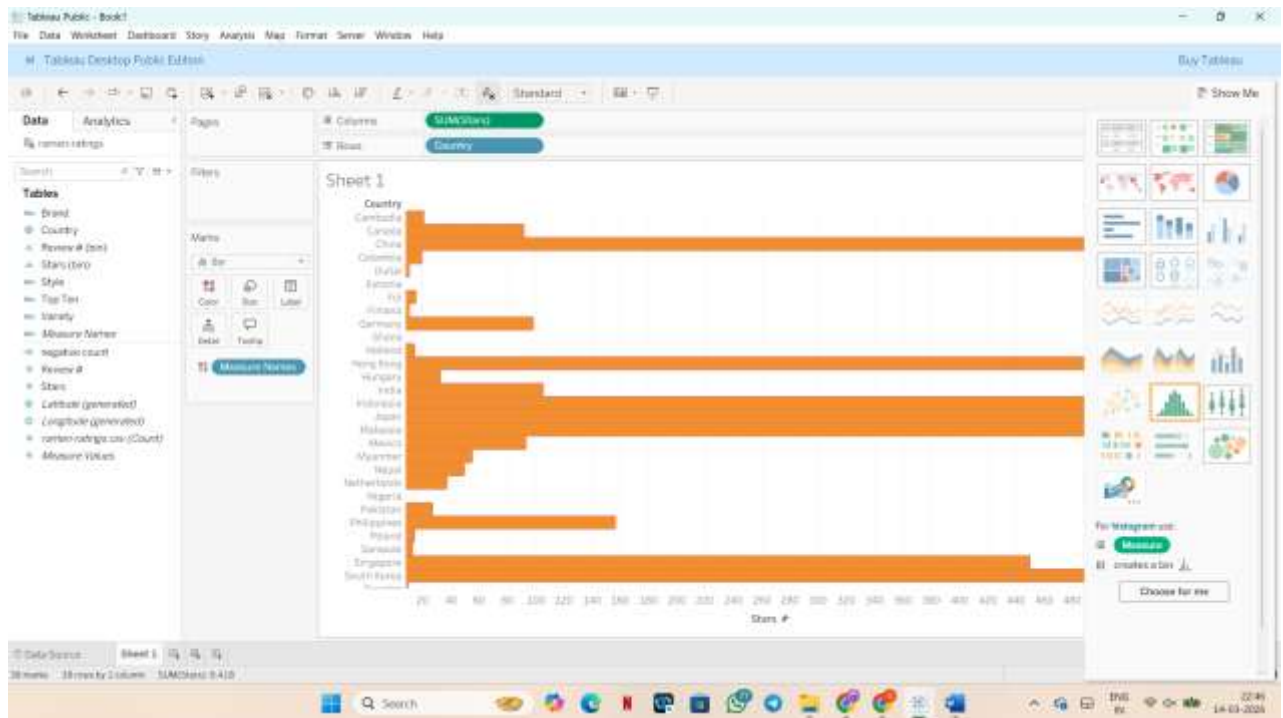


Fig 1: Created Single bar graph Chart with Sum and Country



Fig 2: Created negative count by the above formula

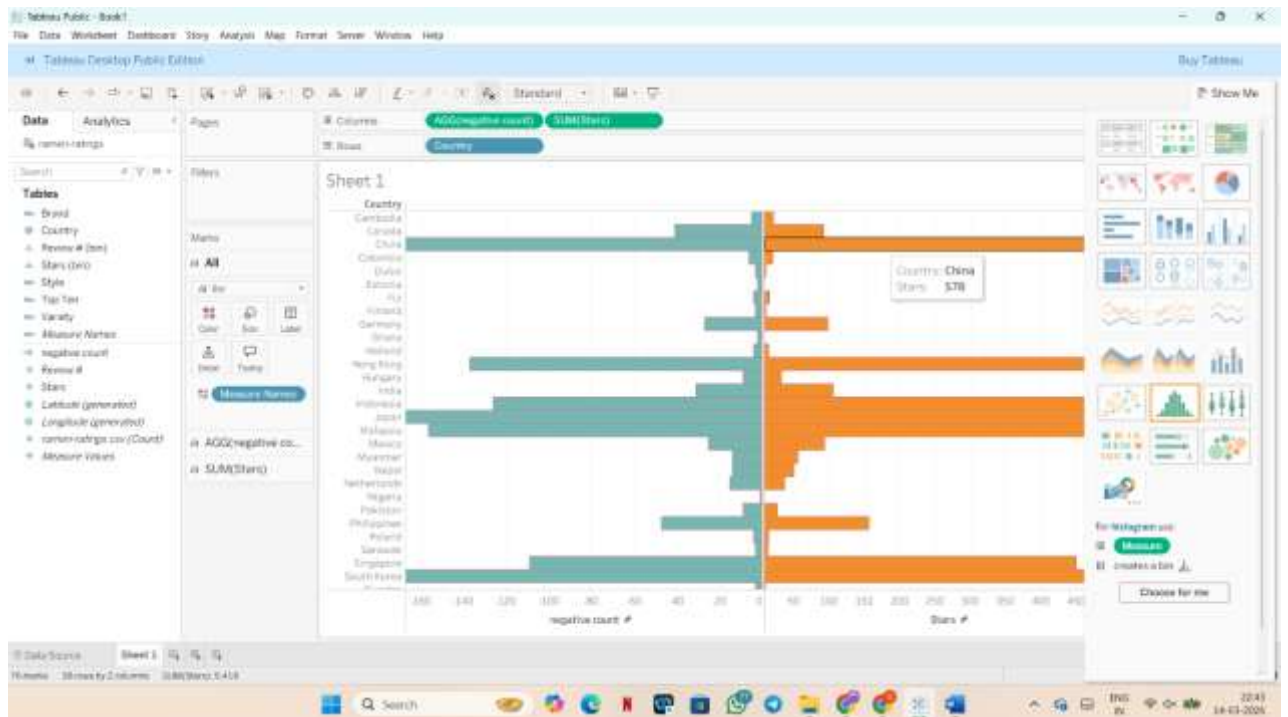


Fig 3: Created butterfly chart with 3 parameters.

Inference:

The butterfly chart helps compare two variables side by side across different categories. It clearly highlights differences in values between the two datasets, making it easier to identify patterns and variations in the data.

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Experiment No: 4

Dt: 14/03/2026

Aim: Establish the relationship between two variables of a data set using scatter plot on Tableau

Requirements:

- Tableau Desktop
- Operating System: Windows /macOS
- Dataset file: ramen-ratings.csv

Step by Step Implementation:

Step 1: Open Tableau and load the dataset ramen-ratings.csv using Connect → Text File.

Step 2: Ensure the Stars field is numeric and remove any Unrated values if present.

Step 3: Drag AVG(Stars) → Columns to represent the average rating.

Step 4: Drag COUNT(Brand) → Rows to represent the number of ramen products.

Step 5: Drag Country → Detail in the Marks Card so that each point represents a country.

Step 6: Ensure the Marks type is Circle to form the scatter plot, and add labels if required.

Output:

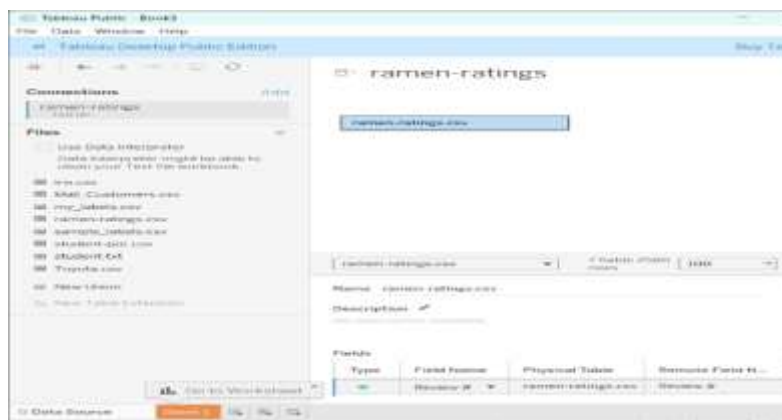


Fig 1: Tableau is launched and dataset is added.

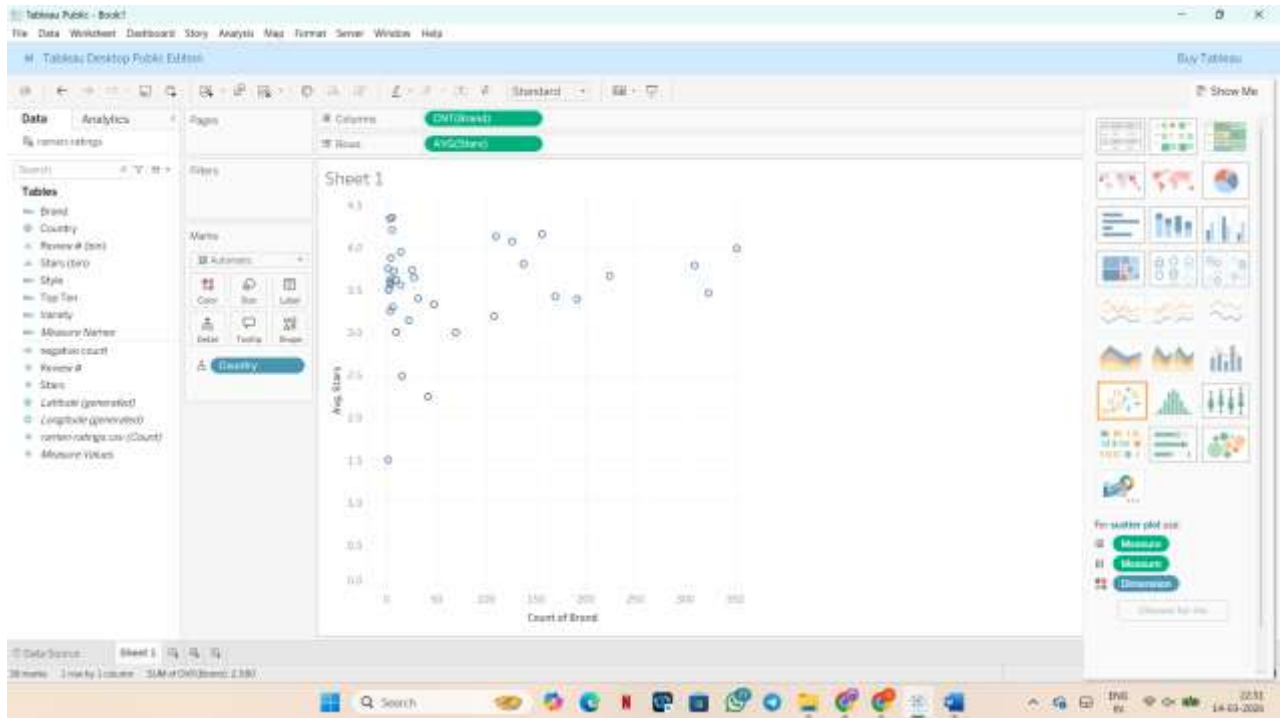


Fig 2: Scatter plot is initiated using count of branch and average of stars

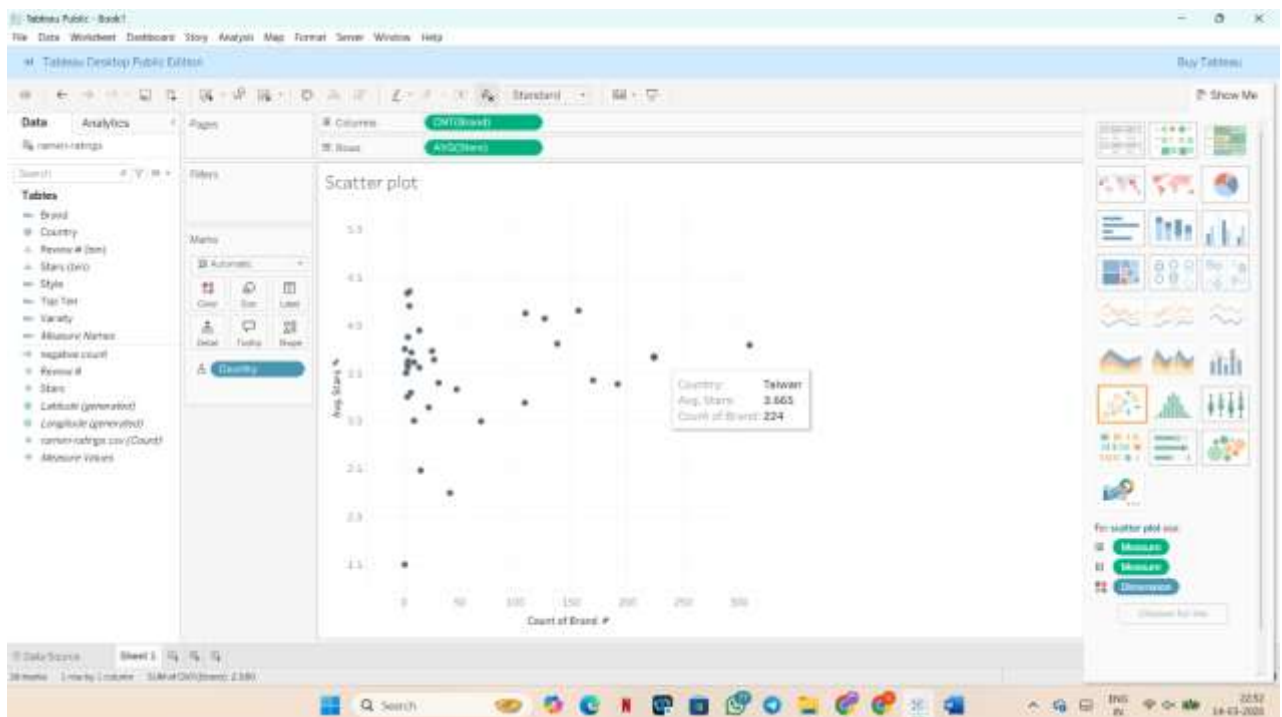


Fig 3: Scatter plot is enhanced with title and colour and shape modifications

Inference:

The scatter plot helps visualize the relationship between two variables. Each point represents a category in the dataset, allowing us to observe patterns, correlations, or trends between the variables.

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Experiment No: 5

Dt: 14/03/2026

Aim: Understand the spread of data with five key using box plot on Tableau

Requirements:

- Tableau Desktop
- Operating System: Windows /macOS
- Dataset file: ramen-ratings.csv

Step by Step Implementation:

Step 1: Open Tableau and load the dataset ramen-ratings.csv using Connect → Text File.

Step 2: Ensure the Stars field is numeric and remove Unrated values if present.

Step 3: Drag Country → Columns and Stars → Rows to create the basic chart.

Step 4: Click Show Me and select Box-and-Whisker Plot to convert the visualization into a box plot.

Step 5: Add labels or colors if required and give the chart a title such as “Box Plot Showing Spread of Ramen Ratings by Country.”

Output:

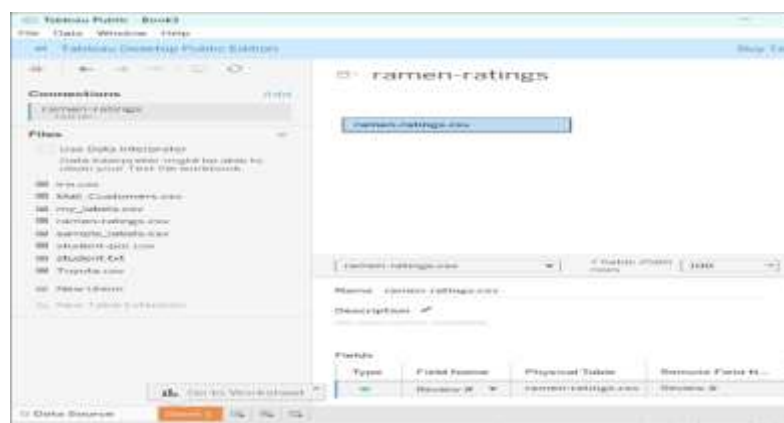
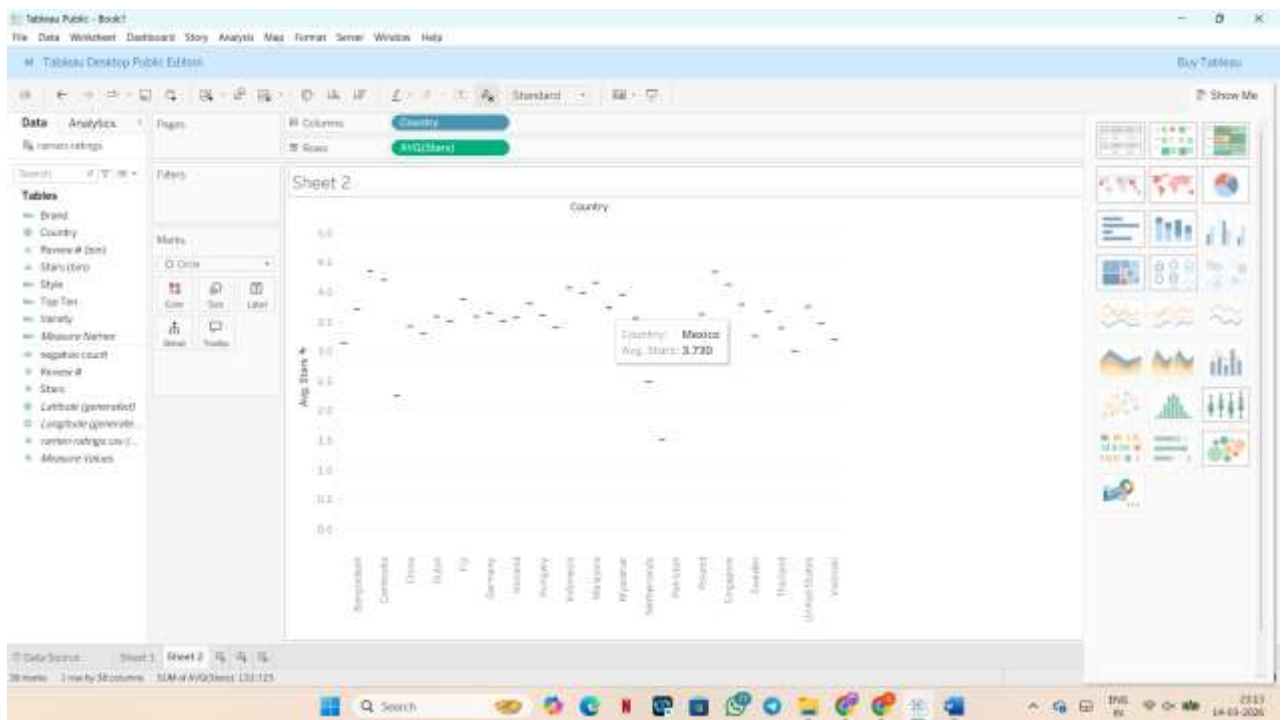
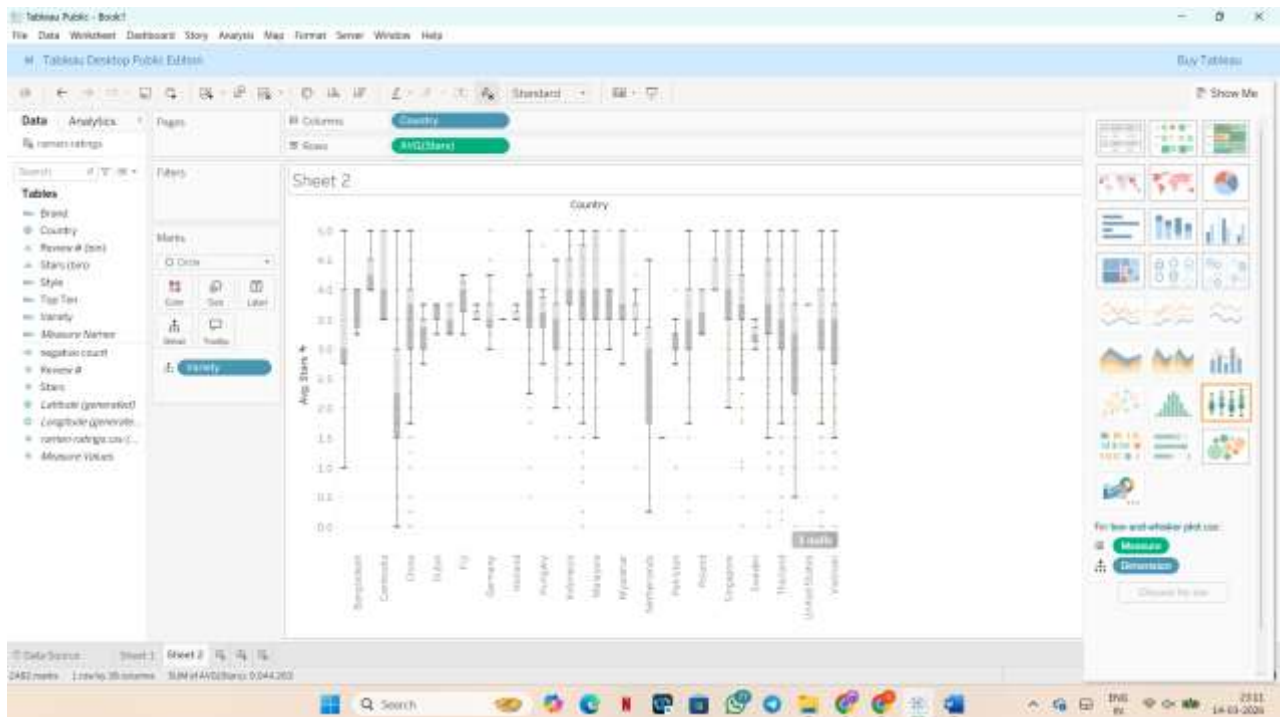


Fig 1: Tableau is launched and dataset is added.



Inference:

The box plot shows the spread and distribution of ratings using the five-number summary: minimum, first quartile (Q1), median, third quartile (Q3), and maximum. It helps identify variation, central tendency, and possible outliers in the dataset.

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Experiment No: 6

Dt: 14/03/2026

Aim: Conduct geographic analysis on sample data using map on tableau

Requirements:

- Tableau Desktop
- Operating System: Windows /macOS
- Dataset file: ramen-ratings.csv

Step by Step Implementation:

Step 1: Open Tableau and load the dataset ramen-ratings.csv using Connect → Text File.

Step 2: In the Data Pane, right-click the Country field and set Geographic Role → Country/Region.

Step 3: Double-click Country to automatically generate a world map visualization.

Step 4: Drag COUNT(Brand) or Number of Records → Color or Size to show the number of ramen products by country.

Step 5: Add AVG(Stars) → Color or Tooltip to visualize the average ratings geographically.

Step 6: Add labels and give the map a title such as “Geographic Analysis of Ramen Ratings by Country.”

Output:

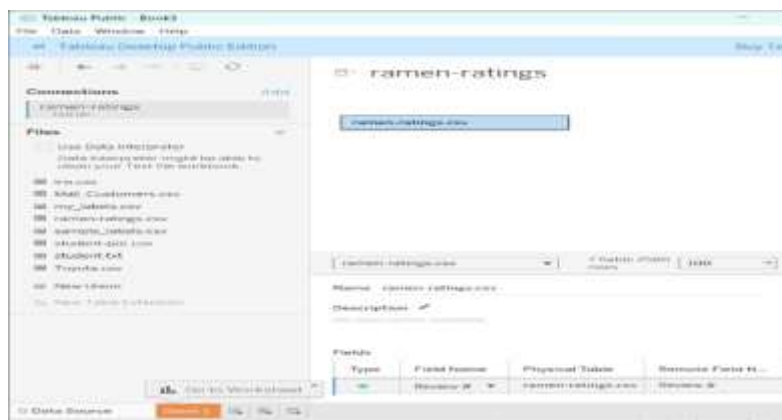


Fig 1: Tableau is launched and dataset is added.

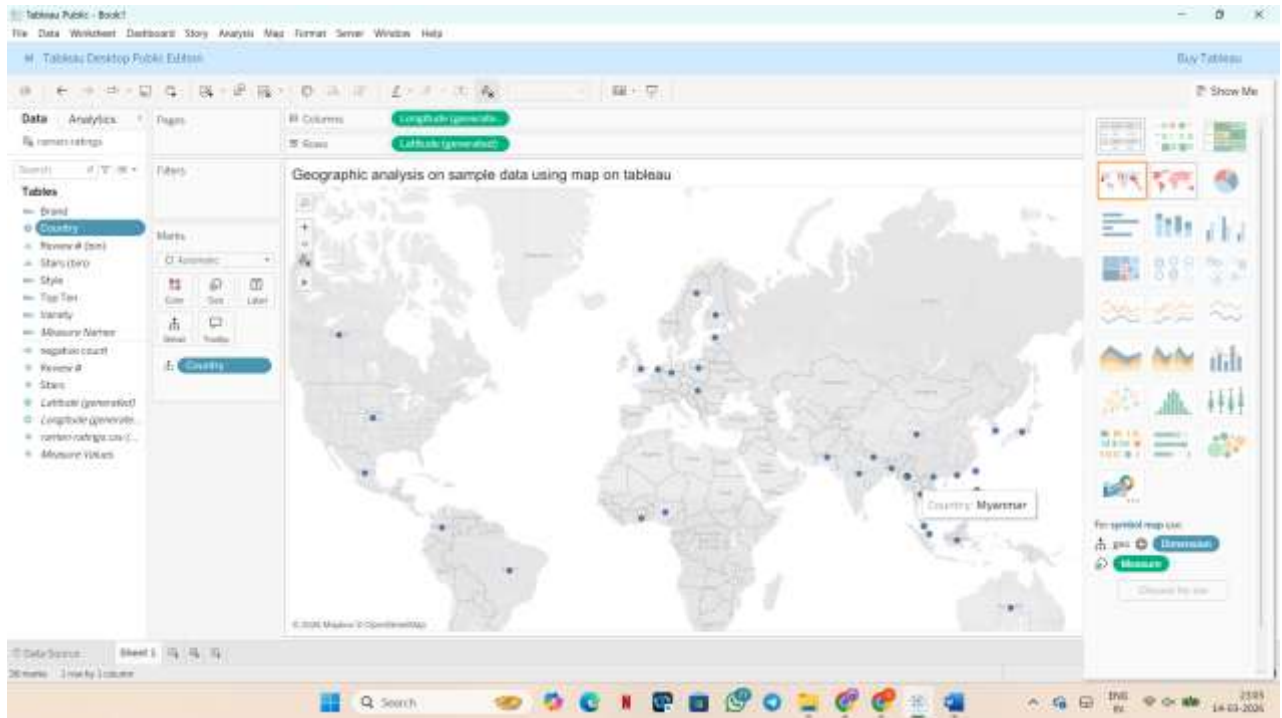


Fig 2: Countries are noted in the created world map

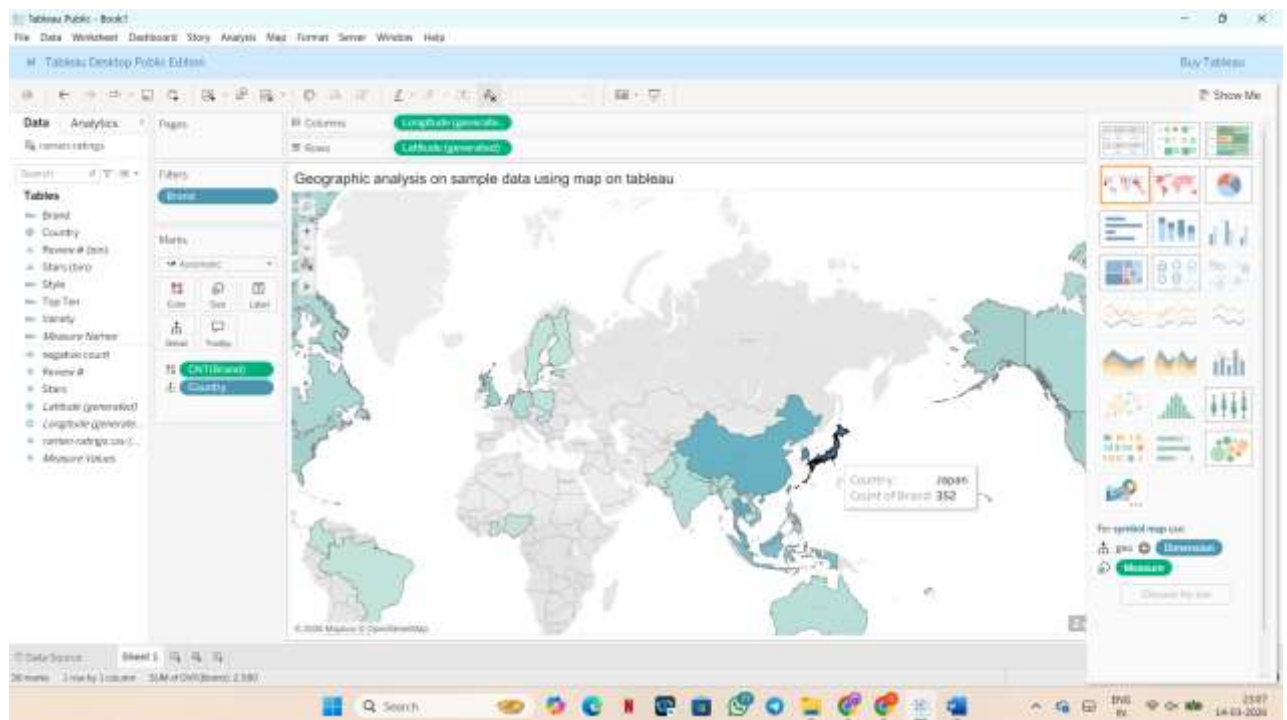


Fig 3: Countries are highlighted

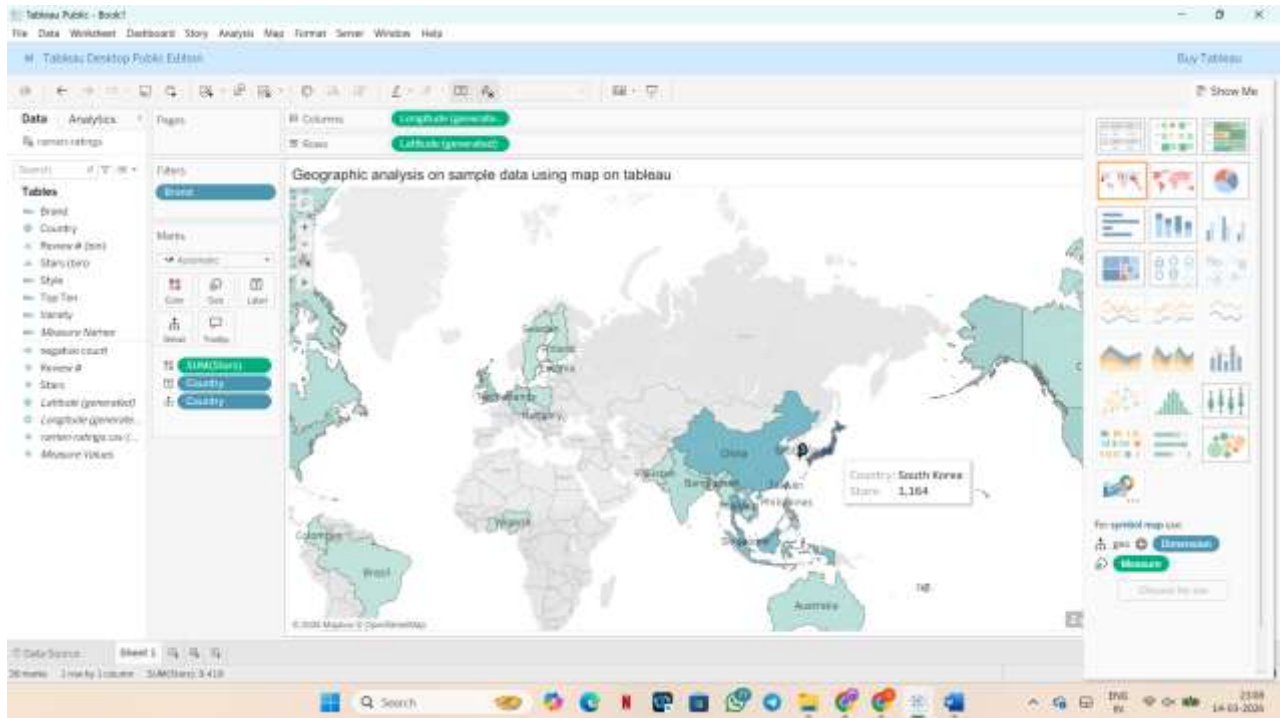


Fig 4: Countries marked and star rating labelled

Inference:

The geographic map visualization shows the distribution of ramen products across different countries. It helps identify regions with higher product counts and higher average ratings, making it easier to observe spatial patterns and geographic trends in the dataset.

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